

Quiz 2 - Part 1: Discrete Mathematics (CSE121)

Answers and rubric

Date: 15-09-2025

Maximum marks: 30

Instructions. Please read the following points carefully.

- Mention your names, roll numbers, room number, and the part number of the quiz in your answer sheets. If we find any details missing, we will not grade the answer sheet.
- Do not use your mobile phones, tablets, laptops, books, notebooks, or any other material during the exam.
- If you are found copying, the matter will be immediately reported to the Disciplinary Action Committee (DAC). If you have any questions, please ask the instructor or TAs.
- This exam is designed for 50 minutes.

1 Problem Set

Question 1.1. *Prove the following only using the properties of integers (given at the end). You should mention which rule you are using at every step.*

1. (2 marks). Suppose $1 = 0$. Prove that $2 = 0$.

Answer + Rubric. We are given that $1 = 0$. On adding 1 to both sides, we get $2 = 1$. (It was shown in the third tutorial that using the 12 properties of integers, we can do the following: for every integers a, b, c , if $a = b$ then $a + c = b + c$. If you mentioned it but not proved, no mark will be deducted. If you did not mention this, 0.5 marks will be deducted.) Now, use the fact that $1 = 0$ and we get $2 = 0$. If it is proved using any other way, then every step of that proof must follow from the 12 properties of integers and appropriate property should be mentioned at every step. The marks will be assigned accordingly.

Remark. Many students have simply multiplied both sides by 2. In this case, they must have shown that we can prove the following using the 12 properties of integers: for every integers a, b, c , if $a = b$ then $c \cdot a = c \cdot b$. If this proof is missing then 1 marks will be deducted.

2. (4 marks). Is the following true: if $1 = 0$ then for every integer n , $n = 0$? Prove your answer. You can use any proof methodology we have covered so far.

Answer + Rubric. This is true. Let n be an integers. Then, there are two cases: either n is odd or it is even. If n is odd then it looks as $n = 2k$ for some integer k , otherwise it

looks as $n = 2m + 1$ for some integer n . We have proved in the first part that $2 = 0$. Then, using the identity rule $a \cdot 0 = 0$ for every integer a , we get that $2k = 2m = 0$. Thus, if n is even then $n = 0$ and if n is odd then $n = 1$. Now, we use the fact that $1 = 0$, which implies $n = 0$. If it is proved this way, one mark will be given to identify two cases, one mark will be given to invoke the previous part (or you can also prove it freshly), one mark should be given for mentioning the identity rule, and one mark should be given for other details. If it is proved using any other way, then every step of that proof must follow from the 12 properties of integers and appropriate property should be mentioned at every step. The marks will be assigned accordingly.

Remark. Once again, many students have simply written that we multiply n on both sides of $1 = 0$ and get $n = 0$, without proving why this is true using the 12 properties of integers only. If such a proof is missing, 2.5 marks will be deducted.

3. (4 marks). Let a, b be integers. If a and b^2 are odd, then prove that $a + b$ is even.

Answer + Rubric. We are given that a, b^2 are odd integers. We have already shown in class that if b^2 is odd then b is odd. If a proof of this is not included then 2 marks will be deducted. As a, b are odd they look like $a = 2m + 1$ and $b = 2n + 1$ for some integers m, n . Then, $a + b = 2m + 1 + 2n + 1$. Now, by using the commutativity of $+$ and distributivity, we get $a + b = 2(m + n + 1)$. Hence, $a + b$ is even. If one of these rules is not mentioned, 0.5 marks will be deducted and if both are missing, 1 mark will be deducted. If it is proved using any other way, then every step of that proof must follow from the 12 properties of integers and appropriate property should be mentioned at every step. The marks will be assigned accordingly.

Question 1.2. Prove the following. You are only allowed to use the properties of integers given at the end of this document. You should mention which rule you are using at every step.

1. (6 marks). Let a, b be natural numbers. If $a \cdot b = 1$ then prove that $a = 1, b = 1$. You can use any proof methodology covered so far.

Answer+Rubric. There are multiple ways to prove this. I am showing a hybrid proof based on proof by contraposition and proof by cases. One objective is to give you another example of a hybrid proof. If it is proved using any other way, then every step of that proof must follow from the 12 properties of integers and appropriate property should be mentioned at every step. The marks will be assigned accordingly.

Let p, q, r be propositions $a \cdot b = 1, a = 1, b = 1$, respectively. We want to prove $p \implies (q \wedge r)$. This is equivalent to $\neg(q \wedge r) \implies \neg p$, which is equivalent to $(\neg q \vee \neg r) \implies \neg p$. If someone has proved it using contraposition and have not shown this, 1 mark will be deducted.

Suppose $(\neg q \vee \neg r)$ is true, which is the same as $a \neq 1$ or $b \neq 1$. We assume without loss of generality that $a \neq 1$. If not, $b \geq 1$, in which case, we replace a with b in proof below.

We know that a, b are natural numbers. As $a \neq 1$, there are two possible cases: either $a = 0$ or $a \geq 2$. If $a = 0$ then, by the identity rule, $a \cdot b = 0$. Hence, $a \cdot b \neq 1$ and $\neg p$ is true. (If $a = 0$ is not considered in the answer 1 mark will be deducted and if it is considered but the identity rule is not mentioned, 0.5 marks will be deducted.)

Now, $a \geq 2$. In other words, a can be written as $m + 1$ for some natural number m such that $m \geq 1$. On substituting $a = 1 + m$ in $a \cdot b$, we get

$$a \cdot b = (1 + m) \cdot b.$$

By using distributivity, we get

$$a \cdot b = 1 \cdot b + m \cdot b.$$

As $b \neq 0$ and $m \neq 0$, by the no zero divisor property, we get that $1 \cdot b \neq 0$ and $m \cdot b \neq 0$. As m, b are natural numbers, this implies that $1 \cdot b \geq 1$ and $m \cdot b \geq 1$. This means there exist natural number x, y such that $1 \cdot b = 1 + x$ and $m \cdot b = 1 + y$. On substituting these in the above equation and using commutativity, we get

$$a \cdot b = 2 + x + y.$$

As x, y are natural numbers, $2 + x + y \geq 2$. This shows $a \cdot b \neq 1$. If some one has directly written that $a \cdot b > 1$ in this case without proving it using only the 12 properties of integers, they will only get 1 mark out of 4. Others will get marks according to their answer. This proves that $\neg p$ is true when $a \geq 2$. Hence, $\neg p$ is true in both cases. This completes the proof.

2. (4 marks). Let a, b be natural numbers. If $a \mid b$ and $b \mid a$ then prove that $a = b$.

Answer + Rubric. Given $a \mid b$ and $b \mid a$, where a, b are natural numbers. It means there exist natural numbers c, d such that $b = a \cdot c$ and $a = b \cdot d$. First suppose $a = 0$, then we get $b = 0$. Now suppose $b = 0$, which implies $a = 0$. Hence, the claim is proven when either $a = 0$ or $b = 0$. If this is not mentioned, 1 mark will be deducted. Henceforth, $a \neq 0$ and $b \neq 0$. On substituting $a = b \cdot d$ in $b = a \cdot c$, we get

$$b = (b \cdot c) \cdot d = b \cdot (c \cdot d) \quad (\text{Associativity of multiplication}).$$

Using the additive inverse and distributivity, the above equation can be re-written as

$$b(1 - c \cdot d) = 0.$$

By the non-zero divisors property, we get that either $b = 0$ or $1 - c \cdot d = 0$. We know that $b \neq 0$. Hence, $1 - c \cdot d = 0$. On adding the additive inverse of $c \cdot d$ to this equation, we get $c \cdot d = 1$. (Four properties of integers have been used here. 0.5 marks will be awarded for each of these properties and for missing any, 0.5 marks will be deducted.) Now, we use the proof of the first part to show that $c = 1$ and $d = 1$. (If this is not mentioned or proved separately, 1 mark will be deducted). Then, this implies $a = b$. If it is proved using any other way, then every step of that proof must follow from the 12 properties of integers and appropriate property should be mentioned at every step. The marks will be assigned accordingly.

Question 1.3. 1. (5 marks). Prove that the sum of a rational number and an irrational number is irrational.

Answer + Rubric. Already done in class. 1 mark will be given to mention that we assume that the given proposition is false. 4 marks will be given to arrive at a contradiction that an irrational number equal a rational number. These 4 marks will be given depending on the steps written in the answer and the steps will be compared to the proof done in class.

2. (2 marks). Assume that the following statements are true. What will be the conclusion of the following statements? Justify your answer.

(a) If a function is differentiable, then it is continuous.

(b) If a function is continuous, then it is bounded on a closed interval.

(c) $f(x)$ is differentiable.

Answer + Rubric. This is proved using Modus Ponens. Let p, q, r, s be the propositions ‘a function is differentiable’, ‘a function is continuous’, ‘a function is bounded on closed interval’, and ‘ $f(x)$ is differentiable.’ We are given that $p \implies q, q \implies r, s$ are true. Then by Modus Ponens, we first get that $f(x)$ is differentiable and then we get that $f(x)$ is bounded on a closed interval. One mark will be given to mention Modus Ponens and 0.5 marks will be given to each of the two conclusions we get.

3. (3 marks). Prove that an integer n is odd if and only if $n + 1$ is even. You are only allowed to use the properties of integers given below.

Answer + Rubric. There are two parts here: if n is odd then $n + 1$ is even and if $n + 1$ is even then n is odd. The proofs of these parts are straightforward and use the definitions of odd and even integers. 1.5 marks will be given to every part. One mark will be deducted for not using the definitions of odd and even integers.

A Important properties of integers

The following properties hold for every integers a, b, c .

1. **Closure under addition and multiplication:** $a + b, a \cdot b$ are integers.
2. **Associativity of addition:** $a + (b + c) = (a + b) + c$.
3. **Associativity of multiplication:** $a \cdot (b \cdot c) = (a \cdot b) \cdot c$.
4. **Commutativity of addition:** $a + b = b + a$.
5. **Commutativity of multiplication:** $a \cdot b = b \cdot a$.
6. **Distributive properties:** $a \cdot (b + c) = a \cdot b + a \cdot c$ and $(a + b) \cdot c = a \cdot c + a \cdot b$.
7. **Identities:** $0 \neq 1, a + 0 = a, a \cdot 0 = 0$, and $a \cdot 1 = a$.
8. **Additive inverses:** There exists a unique integer m such that $m := -1 \cdot a$ and $a + m = 0$. The inverse of a is denoted $-a$.
9. **Subtraction:** $a - b$ is defined as $a + (-b)$.
10. **Cancellation:** If $a \cdot b = a \cdot c$ and $a \neq 0$ then $b = c$.
11. **No zero divisors:** If $a \cdot b = 0$ then either $a = 0$ or $b = 0$.
12. **Signed product:** $-a \cdot b = (-a) \cdot b = a \cdot (-b)$.